

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

November – December 2011

CL 203 Fluid Mechanics

Date: 1.12.2011, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1** (a) Define: Ideal Fluid, Real Fluid and Viscosity [03]
- (b) Find the kinematic viscosity of an oil having density 980 kg/m^3 . The shear stress at a point in oil is 0.26 N/m^2 and velocity gradient at that point is 0.25 per second. [02]
- (c) Define Manometer and give classification of the same. [02]
- Q - 2** (a) State and prove Pascal's Law [04]
- (b) A rectangular pontoon is 6 m long, 3 m wide and 1.20 m high. The depth of immersion of the pontoon is 0.80 m in sea water. If the center of gravity is 0.60 m above the bottom of the pontoon, determine the meta-centric height. The density of sea water is 1025 kg/m^3 . [03]
- (c) Define Forced Vortex Flow and Free Vortex Flow giving examples. [07]
- An open circular tank 30 cm diameter and 100 cm long contains water upto a height of 60 cm. The tank is rotated about its vertical axis at 280 r.p.m. Find the depth of parabola formed at the free surface of water.

OR

- Q - 2** (a) Explain the following terms: [03]
- (i) Path Line, (ii) Laminar flow, (iii) One-dimensional flow.
- (b) Define: Local Acceleration. [04]
- The two velocity components of fluid flow are given below. Determine the third component of velocity such that the continuity equation is satisfied.
- $$u = x^2 + y^2 + z^2; v = xy^2 - yz^2 + xy.$$
- (c) Derive an expression for the total pressure and position of centre of pressure on a [07]
- plane surface immersed vertically in a liquid.
- Q - 3** (a) The stream function for a two-dimensional flow is given by $\Psi = 8xy$. Calculate the [05]
- velocity at the point (4, 5). Find the velocity potential function ϕ .

- (b) Derive an expression for discharge through fully submerged orifice. [05]
- (c) A horizontal venturimeter with inlet diameter 30 cm and throat diameter 15 cm is used to measure the flow of oil of specific gravity 0.8. The discharge of oil through venturimeter is 50 liters/sec. Find the reading of the oil-mercury differential manometer. Take $C_d = 0.98$. [04]

OR

- Q - 3 (a) Derive the continuity equation for three dimensional flow. [05]
- (b) Give classification of notches. [05]
- Determine the height of a rectangular weir of length 5 m to be built across a rectangular channel. The maximum depth of water on the upstream side of the weir is 1.5 m and discharge is 1500 liters/sec. Take $C_d = 0.6$ and neglect end contraction.
- (c) Classify Mouthpiece based on their shapes. [04]
- A convergent-divergent mouthpiece is fitted to the vertical side of a tank, containing water. It's throat and exit diameters are 6 cm and 12 cm respectively. Calculate the maximum head of water for steady flow. The maximum vacuum pressure is 9 m of water and tank atmospheric pressure = 10.3 m of water.

SECTION – II

- Q - 4 (a) State Bernoulli's equation. [04]
- Water flows in a circular pipe. At section -1 the diameter is 0.4 m, the static pressure is 270 kPa gauge, the velocity is 2.5 m/s. & the elevation is 12 m above ground level. The elevation at a section downstream is zero and the pipe diameter is 0.2 m. Determine the gauge pressure at the downstream section.
- (b) Define momentum correction factor & enlist application of momentum equation. [03]
- Q - 5 (a) Derive an expression of power transmission through pipe. Also derive the equation for maximum transmission of power. [05]
- (b) Enlist the methods of dimensional analysis. Describe any one in detail. [04]
- (c) Define any three dimensionless numbers. [05]
- Water is flowing through a pipe of diameter 30 cm at a velocity of 4 m/s. Find the velocity of oil flowing in another pipe of diameter 10 cm, if the condition of dynamic similarity is satisfied between the two pipes. The viscosity of water & oil is given as 0.01 poise & 0.025 poise. The specific gravity of oil = 0.8

OR

- Q - 5 (a) What is compound pipe? Write equation for head loss when pipes are connected in series? [02]

- (b) Calculate the discharge through a pipe of diameter 200 mm when the difference of pressure head between the two ends of a pipe 500 m apart is 4 m of water. [03]
Use $f = 0.009$ in the Darcy-Weisbach formula.
- (c) A partially submerged bogy is towed in water. The resistance R to its motion depends on the density ρ , the viscosity μ of water, length l of the body, velocity v of the body & acceleration due to gravity g . Express the functional relationship between these variables and resistance R . [06]
- (d) Explain method of selecting repeating variables for dimensional analysis. [03]
- Q - 6 (a) A supersonic aircraft flies at an altitude of 1.8 km, where the temperature is 5°C . Derive the speed of the aircraft if its sound is heard 5 seconds after it passes over the head of an observer. Take $R = 287 \text{ J/kg K}$ & $\gamma = 1.4$. [03]
- (b) Calculate the stagnation pressure and temperature on the stagnation point on the nose of a plane, flying with a speed of 1600 km/hr at an altitude of 10 km above sea level through still air having a pressure and density 2.4 N/cm^2 and 0.5 kg/m^3 respectively. Use $R = 287 \text{ J/kg K}$ & $\gamma = 1.4$. [03]
- (c) Describe in detail the momentum thickness, (θ) . [03]
- (d) A laminar flow is observed in a pipe of diameter of 200 mm. The maximum velocity is 1.5 m/s. Find the mean velocity and the radius at which mean velocity occurs. Also calculate the velocity at 4 cm from the wall of the pipe. [05]

OR

- Q - 6 (a) A supersonic plane moves with a Mach number 3 at an elevation of 1000 m. The atmospheric temperature at given height is 2°C . Calculate the time elapsed between the instant the aircraft was directly overhead of an observer & the instant the observer feels the sound due to aircraft for the stationary position of observer. Take $R = 287 \text{ J/kg K}$ & $\gamma = 1.4$ [03]
- (b) Derive the Bernoulli's energy equation for one-dimensional compressible adiabatic flow. [03]
- (c) Determine (a) the pressure gradient, (b) the shear stress at the two horizontal parallel plates & (c) the discharge per meter width for the laminar flow of oil with a maximum velocity of 1.5 m/s between two horizontal parallel fixed plates which are 80 mm apart. Take viscosity of oil = 1.962 N-s/m^2 . [03]
- (d) Derive an expression for the velocity distribution for viscous flow through a circular pipe. [05]
